Reg. No. | | | | | | | | | | | [18MSCH3H01]

THIRD SEMESTER UNIVERSITY EXAMINATION, DECEMBER 2021

INORGANIC CHEMISTRY-III

Time : 03 Hours **Maximum Marks:70**

Instructions: 1. Question 1 is compulsory, carries 10 marks.
2. Questions 2 to 7 carries 15 marks each, answer any FOUR.

1. Answer any FIVE of the following questions. **5 x 2 = 10**

a. What are the factors favouring the outer sphere electron transfer reaction?

b. The rate of aquation of $[CoCl_2(Me_4en)_2]^+$ is 1000 times faster than that of $[CoCl_2(en)_2]^+$, Explain.

c. Define metallocene with an example.

d. Differentiate homogeneous and heterogeneous catalysts with examples.

e. What is a skeletal electron pair? Explain it using [BH] unit.

f. Define photochemistry.

2. a. What is the *trans* effect? Explain with relevance to square planar complexes.

b. Explain the Marcus theory of electron transfer with potential energy diagram.

c. Explain the inner sphere reaction mechanism with suitable examples. **(5+5+5)**

3. a. Discuss experimental evidence for dissociative mechanism pathway in substitution reactions of octahedral metal complexes.

b. Discuss in detail the metal ion and ligand specific factors affecting stability of metal complexes with examples.

c. What is S_N^1 CB (conjugate base) mechanism? Deduce the rate equation for the base hydrolysis of $[Co(NH_3)_5Cl]^{2+}$, which proceeds *via* S_N^1 CB mechanism. **(5+5+5)**

4. a. Explain the structure and bonding involved in metal nitrosyl complexes.

b. Give an account of synthesis and structural features of *ansa*-metallocenes. **(10+5)**

5. Explain in detail the following:

a. Oxidative addition and reductive elimination

b. β-hydride elimination reaction

c. Carbonylation with suitable organometallic catalyst. **(5+5+5)**

6. a. What is metal-boride compound? Write synthesis, structure, and reactivity of $[HFe_4(CO)_{12}BH_2]$ iron-boride cluster.

b. Predict the structure of the following boranes using PSEP (polyhedral skeletal electron pair) theory: i) B_4H_{10} and ii) B_5H_{11}

c. Define boranes, carboranes, metallaboranes and metallacarboranes. Illustrate how they are interrelated to each other by flow-chart diagram. **(5+5+5)**

7. a. Discuss the optical transitions in semiconductors.

b. Describe the bandgap energy formation in semiconductors. **(10+5)**

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